

PalaMulti Project: Global Lexical Integration

Optimization Progress Across 22 Languages

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The Integration Pipeline: From Raw to Refined

- **Phase 1 (Raw Mapping):** Initial dictionary-form matching. Challenges included inflection mismatches, script segmentation, and polysemy.
- **Phase 3 (Final Production):** Application of the pass3 engine.

Key Optimization Steps

- 1 **Normalization:** Handling Unicode and script-specific variations.
- 2 **Morphological Recovery:** Using tools like *SudachiPy* (JA) and *Pyidaungsu* (MY).
- 3 **Sense Reduction:** Eliminating "orphan" candidates to reach 100% technical coverage.

The spaCy Effect: Comparison of Success Rates

- **String Match (Baseline):** Failed on inflected forms (e.g., *eating* vs. *eat*).
- **spaCy Optimized:** Contextual lemmatization + PhraseMatcher.

Language	String Match %	spaCy Optimized %	Improvement
English (en)	92.1%	100.0%	↑ 7.9%
German (de)	88.4%	100.0%	↑ 11.6%
Japanese (ja)	85.0%	100.0%	↑ 15.0%
Chinese (zh)	81.3%	100.0%	↑ 18.7%
Vietnamese (vi)	78.2%	100.0%	↑ 21.8%

Without spaCy

High failure in Asian languages due to missing word boundaries.

With spaCy + Pass 3

100% technical mapping via automated lemmatization.

Mapping Evolution: East & Southeast Asia

Language	Phase 1 (Raw)	Phase 3 (Final)	Improvement
Japanese (ja)	85%	100.0%	+15%
Chinese (zh)	81%	100.0%	+19%
Korean (ko)	83%	100.0%	+17%
Vietnamese (vi)	78%	100.0%	+22%
Thai (th)	74%	100.0%	+26%
Burmese (my)	72%	100.0%	+28%
Khmer (km)	75%	100.0%	+25%
Tagalog (tl)	88%	100.0%	+12%
Malay / Indonesian	86%	100.0%	+14%

Finding

The highest gains (+28%) were achieved in Southeast Asian languages where word segmentation was the primary barrier in Phase 1.

Mapping Evolution: Europe & Americas

Language	Phase 1 (Raw)	Phase 3 (Final)	Improvement
English (en)	92%	100.0%	+8%
German (de)	88%	100.0%	+12%
French (fr)	90%	100.0%	+10%
Spanish (es)	91%	100.0%	+9%
Portuguese (pt/pb)	89%	100.0%	+11%
Russian (ru)	82%	100.0%	+18%

Analysis

High-resource languages started with high baselines, but the pass3 engine was critical for resolving complex verbal inflections in Russian and German.

Mapping Evolution: ME, Central & South Asia

Language	Phase 1 (Raw)	Phase 3 (Final)	Improvement
Arabic (ar)	79%	100.0%	+21%
Urdu (ur)	76%	100.0%	+24%
Turkish (tr)	84%	100.0%	+16%
Mongolian (mn)	80%	100.0%	+20%
Assamese (as)	73%	100.0%	+27%

Script Diversity

RTL (Right-to-Left) and Brahmic scripts required specific normalization layers to bridge the gap between TUFS and WordNet identifiers.

Technical Coverage vs. Semantic Accuracy

- **Definition of 100%:** Every entry in the final production files is successfully linked to a non-orphan Synset ID.
- **Verification:** Automated validation of XML schema and attribute presence across 10,500+ records.

Manual Verification (Future Work)

While technical coverage is complete, the next phase involves human-in-the-loop validation to ensure the "Best-Fit" selection correctly handled high-polysemy words.

Thank you for your attention.
Questions?